Model & Data Card

ForceCRM | Local profile discovery for Greek foodservice businesses

Inputs and intended use

A sales representative uploads customer VAT numbers and revenue. ForceCRM groups customer venues, describes statistically supported traits in Greek and ranks one prospective venue per non-customer VAT. A person verifies suitability before contacting a business.

The supplied GEMI extract contains 24,424 venues, 23,945 distinct VATs and 3,736 Greek activity descriptions. GPS coverage is 96.0%. All 300 portfolio VATs matched. The competition scope covers businesses established during 2020-2024 with a primary activity in division 56.

Profiles in the supplied portfolio

Profile	Customers	Revenue share	Stability[1]	Fit confidence[2]
ICP-01	126	25.8%	93.0%	55.5%
ICP-02	74	27.4%	88.4%	57.8%
ICP-03	100	46.8%	91.4%	68.3%

[1] Mean membership Jaccard across 20 VAT bootstraps with matched profiles and fixed cluster count. [2] Mean relative fit to the profile radius. Sales conversion requires separate outcome data.

Internal retrieval evaluation

Rank cutoff	ForceCRM recall	NACE baseline	Difference
Top 500	25.0%	1.7%	+23.3 percentage points
Top 2,000	55.0%	8.3%	+46.7 percentage points

Seed 2026 held out all venues of 60 customers before fitting on the other 240. The baseline ranks primary-code frequencies in the training portfolio. The organizer retains its separate hidden evaluation. A single development split gives a limited estimate of future sales performance.

Grounding, oversight and data rights

Fisher tests compare trait prevalence at VAT level. Global Benjamini-Hochberg correction, at least five supporting VATs and lift of at least 1.2 limit chance findings. Clustering on the same data makes these associations exploratory. Missing GPS, self-declared activities and multiple branches can distort similarity.

GEMI competition data. Geocoding and venue nodes © OpenStreetMap contributors (ODbL), via Geofabrik. ForceCRM divided ELSTAT 2024 hotel nights by 2021 census residents. ELSTAT bears no responsibility for this transformation. Map outline: Natural Earth (public domain). Optional live registry-status check: ΓΕΜΗ Open Data API (ODC-BY 1.0), opt-in, sends prospect VATs only. Sales research only.

Model selection and cost

Measurements on Apple A18 Pro with 8 GiB RAM, CPU/GPU as scheduled by Ollama

Greek activity embeddings

Multilingual E5-small encodes activity descriptions into 384 dimensions. The US-based Microsoft Research E5 team trained the model on multilingual web and translation pairs, then supervised retrieval data. The benchmark encoded 75 Greek foodservice descriptions and checked top-five retrieval against their four-digit activity families.

Embedding model	Family hit@5	ms / description	Process RSS[3]
e5-small	93.2%	17.6	300 MiB
e5-base	93.2%	87.2	330 MiB
multilingual-minilm	89.0%	15.9	381 MiB

[3] Sampled after encoding and inclusive of Python allocations. E5-small matched E5-base on this code-family proxy and required less time and memory. Expert judgments would provide a direct test of semantic relevance.

Local Greek narrative drafting

Mistral AI in France developed Ministral 3 3B Instruct. The Q4_K_M package occupies approximately 3 GB. Full corpus provenance remains undisclosed. Each profile fact is first written as a Greek template sentence. The LLM then rewrites one sentence at a time under a JSON schema. Numbers, percentages and names are checked against the source fact, and misleading wording is rejected. A sentence that fails twice keeps its verified template. Each displayed sentence links to its source statistic and records its author. Every description ends with a computed sentence stating profile quality, bootstrap stability and fit confidence. Generated descriptions carry an AI label.

LLM	LLM-written sentences[4]	Time per profile	Ollama memory
ministral-3:3b	11 / 18 (61%)	50.0 s	2.33 GiB
qwen3:1.7b	18 / 18 (100%)	32.2 s	1.13 GiB
gemma3:1b	17 / 18 (94%)	16.8 s	0.82 GiB

[4] Sentences that passed every deterministic check on the first or second attempt; the rest kept their verified template. All models receive the same profiles, prompt and checks. This measures grounded rewriting, not fluency. Human Greek fluency ratings remain uncollected.

In the submitted run Ministral wrote 100% of fact sentences and took 24-27 seconds per profile. The pass rate varies between runs with the local Ollama state (benchmark run above); failing sentences always keep their verified template. The full run took 1.6 minutes with cached activity embeddings. Temperature 0, seed 42, top-k 1 and context 2,048 are fixed. Each export records model identity, prompt hash, rejected drafts and source hashes.

Licenses and reproducibility

The distributed Ministral license is Apache 2.0 and allows commercial use under its terms. The E5-small weight repository declares MIT in its metadata and lacks a standalone license file. The official E5 code repository supplies the MIT license. The weight-specific licensing evidence is limited to the publisher’s declaration.

OpenAI Codex assisted implementation, documentation and testing. Claude Code (Anthropic) assisted code review, holdout verification, setup fixes, the grounded per-sentence narration and documentation. The repository includes source links, an EU AI Act self-assessment, benchmark commands and the temporary Colab workflow.